

HARIS WAHEED BHATTI

BIM Specialist | BIM Software & Automation Developer

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Date of Birth: 25.12.2001 | Nationality: Pakistani

Open to roles in Islamabad or Dubai; available immediately.

PROFILE

Civil engineer with a European Master's in Building Information Modelling who builds the software that automates BIM work. Published an open-source IFC-to-Neo4j converter on PyPI, built an LLM agent that answers questions directly against IFC models, and owned a federated model's clash-detection workflow across architecture and MEP disciplines under ISO 19650.

PROJECTS

ifctoneo4j – Python package (PyPI) | Python, IfcOpenShell, Neo4j, Cypher 2026

- Open-source library converting IFC models (IFC2X3, IFC4, IFC4X3) into Neo4j property graphs aligned to Linked Building Data ontologies, with no RDF intermediate. Published to PyPI: `pip install ifctoneo4j`.

BIMGraphExplorer – LLM agent for IFC question answering | Python, MCP, Neo4j 2026

- Adaptive querying system exposing four read-only tools over the Model Context Protocol; profiles the graph schema and issues iterative Cypher queries conditioned on prior results.
- Raised answer accuracy by 33 percentage points and source transparency from 12.5% to 99.7% on a fixed model and dataset, through prompt design alone.

Federated BIM Model – Pakistani-Themed Expo, Guimarães | IFC, Autodesk Construction Cloud Oct 2025

- Owned the federated model and clash-detection workflow across architecture and MEP disciplines, validated against Employer's Information Requirements under ISO 19650.

EDUCATION

MSc, European Master in BIM (BIM A+) | UL FGG, Ljubljana, Slovenia Oct 2025 – Jul 2026

- Field of study: Information Management in Construction. Completed the accelerated thesis track, finishing full MSc requirements in 10 months. Final grade 9.27/10.
- Thesis: *Graph-Mediated LLM Exploration for IFC Question Answering: A Benchmark Comparison Against Direct-IFC Approaches*. Supervisor: Prof. Žiga Turk. hdl.handle.net/20.500.12556/RUL-185169
- Applied coursework spanning the BIM delivery chain: scan-to-BIM from point-cloud capture, IFC schema auditing and model remediation, parametric façade optimisation in Dynamo, process modelling and workflow improvement, and 5D quantity take-off and cost estimation.

BE Civil Engineering | NUST, Islamabad, Pakistan. Final grade 3.25/4.00 Sep 2020 – Jun 2024

EXPERIENCE

Software Engineering Fellow | Headstarter AI (remote) 2024

- Built AI-powered applications in Python and JavaScript over a 7-week fellowship, including weekend hackathon projects and a capstone application.

Civil Engineering Intern | Fatima Fertilizer Company, Pakistan July 2023

- Modelled a 6,804 m² rock phosphate storage shed in 3D from 2D drawings, coordinating with stakeholders on design refinement.

SKILLS

BIM & AEC: Revit, AutoCAD, Dynamo, scan-to-BIM (point cloud to model), clash detection, federated models, IFC (IFC2X3, IFC4, IFC4X3), ISO 19650, Employer's Information Requirements, Autodesk Construction Cloud

BIM Software Development: Python, TypeScript, JavaScript, IfcOpenShell, That Open Engine, Fragments, Neo4j, Cypher, knowledge graphs, Git

AI / LLM: Agentic systems, Model Context Protocol (MCP), RAG, LLM evaluation and benchmarking

Languages: English (C1), Slovene (A1, actively learning)

CERTIFICATIONS

Master BIM Software Developer (in progress) | That Open Company Dec 2024 – present

- Building AEC web applications with That Open Engine, Fragments and TypeScript.